more than double to $100-110 billion (Rs 9.6-10.6 lakh crore) by 2030.

The first phase of the mission, unveiled in December 2021, has laid the foundations for this expansion. Under Semicon 1.0, the government approved 12 semiconductor units, with cumulative investments exceeding Rs 1.64 lakh crore. These include three chip fabrication units and nine packaging and testing units. Three companies—Micron, Kaynes and CG Semi—have already begun commercial production; another facility is expected to become operational later this year. More than 105 start-ups are developing indigenous chips, while around 315 universities have started training students using advanced electronic design automation tools.

BUILDING ON EARLY GAINS

Semicon 2.0 seeks to build on these early gains through a broader, six-pronged strategy. Besides attracting more fab plants, the policy aims to deepen India's design ecosystem, encourage domestic production of specialised machinery and materials used by the industry, expand packaging and testing capacity, support advanced research and build a larger pool of skilled talent.

The emphasis in chip design will shift towards creating strategic and commercially viable intellectual property (IP). The government also plans to incentivise companies engaged in the R&D and manufacture of machinery, chemicals, gases and other materials essential for semiconductor production, while research efforts will focus on advanced technologies in collaboration with leading institutions.

Industry experts say the first phase has created momentum, but scaling up will require sustained public support. "The government's push in the form of various incentive schemes has helped, with 12 major plants getting approval," says Vishal Goel, analyst at HSBC Global Investment Research. "However, the outlay compared to other global economies is far less and needs to be increased. Given the huge initial capex and long gestation period required, government support is paramount."

AGENDA FOR SEMICON 2.0

- Deepen the semiconductor design ecosystem
- Incentivise R&D and manufacturing of machines and materials for the industry
- Set up more chip fabrication units
- Strengthen the packaging and testing industry
- Develop advanced technologies through leading R&D centres
- Build a large pool of skilled talent

₹1.27 lakh cr.
Budget outlay for India Semiconductor Mission 2.0 (Semicon 2.0)

₹1.64 lakh cr.
Investments attracted in 12 semiconductor units under Semicon 1.0

India does enter the race with structural advantages, he notes. It is already a global hub for semiconductor design, with an estimated one-fifth of the world's chip design engineers working from global capability centres in Bengaluru, Hyderabad and Noida. As multinational companies pursue a 'China+1' strategy to diversify their supply chains, India has emerged as an alternative manufacturing destination. Moreover, India's rapidly expanding electronics, automotive and telecommunications industries provide a growing domestic market for semiconductors. For instance, under the production-linked incentive (PLI) schemes, the country's electronics production is projected to reach $300 billion (Rs 29 lakh crore) this year. In mobile phones, in particular, domestic production now meets almost the entire local demand.

THE BOTTLENECKS AHEAD

Yet translating these advantages into a globally competitive semiconductor industry will not be easy. A single fabrication plant can require investments of $3-10 billion (Rs 28,900-96,400 crore), while chip manufacturing depends on more than 300 highly specialised materials and equipment inputs, many of which still need to be imported. Together with long gestation periods, this could deter private investment despite government support.

That, says Goel, is why India's biggest near-term opportunity may lie in packaging and assembly. "The problem is that the IP [of advanced technologies] lies with parent companies located in the US or Europe. Design software is costly and raw material supply chains are choppy," he notes. "Therefore, India has an immense opportunity to capture the packaging and assembly part of the value chain as this does not require large amounts of capex or deep talent expertise."

Competition is another hurdle. China, Taiwan, the US and Japan continue to dominate advanced semiconductor manufacturing, while Southeast Asian hubs such as Vietnam and Malaysia are aggressively expanding their own capabilities. India must also overcome structural bottlenecks, including an uncertain tax regime, a shortage of skilled labour at competitive salaries, and the inability to provide uninterrupted power supply—critical for semiconductor manufacturing.

S. Sathish, partner at KPMG India, says Semicon 2.0 will need a strong governance framework to ensure that funding translates into meaningful national capability building. "Dedicated boards should be established for each of the six focus areas, with clear accountability for outcomes, progress tracking and bottleneck resolution." At the apex, a central command centre should monitor milestones, facilitate stakeholder participation and drive coordination across ministries, states, industry and academia, he adds. Only then can India turn its semiconductor ambitions into a globally competitive reality. ■